

The b-Correction and the Vortex Volume

What volume of water corresponds to one b-correction? The monograph proves and numerically verifies the exact div-injection identity: the carrier of the correction is vorticity, and the elementary b-volume is given by the constant $4\pi+2\sqrt{3}$ from the denominator of the correction itself. The merging algebra is derived, the extended cross-check battery H/I/J/K measures every quantity through two independent channels, and the dynamic Hou-Luo experiment shows when corrections add and when they annihilate.
wild8highlander · the navier-stokes-b project

Repository: github.com/wild8highlander/navier-stokes-b

Companion: [research/b_volume_experiment.py](#) · [research/b_volume_extension.py](#) · [research/b_volume_check.jl](#)

$\theta_b = \arcsin(1/(4\pi+2\sqrt{3})) = 3.5765^\circ$ · $V_1 = 4\pi+2\sqrt{3} \approx 16.02 = 6.46\%$ of the box
монитор $Q/(\theta_b \cdot B) = 1 + O(\theta_b)$ · 12 опытов · все PASS · Python + Julia

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1. Introduction: hypothesis, problem, main result

In the navier-stokes-b repository the b-correction lives as a purely mathematical object: the angle $\theta_b = \arcsin(b)$, with $b = 1/(4\pi + 2\sqrt{3}) \approx 0.062381$, through which the velocity vector is rotated pointwise on the grid, after which the field is re-projected by the Leray operator. Everything the standard correction audit measures is dimensionless: the 10^{-12} isometry of the rotation, the injected divergence, the response of $\sup|\omega|$. Water — and any liquid — however occupies a volume, and the repository author put forward a substantive physical hypothesis: **one water vortex carries one b-correction of its own; merging vortices accumulate their corrections and grow without bound; hence the magnitude of the correction is directly tied to the volume of the liquid.**

This monograph turns that hypothesis into a quantitative theory and backs it with numerical experiment. We give an operational definition of the «amount of correction» — the **b-charge** $B = \int |\mathbf{n}_b \cdot \omega| dV$, derive the exact volume formula, answer the main question of which volume corresponds to one correction, establish the merging algebra — when corrections add and when they annihilate — and build the Kolmogorov bridge from dimensionless code units to cubic meters of real water. A separate chapter is devoted to the extended cross-check battery: every quantity of the theory is measured by at least two independent routes — through the vorticity and velocity channels, through L^1 - and L^2 -carriers, on four grid resolutions, and dynamically in the Hou–Luo annihilation experiment.

The central analytic result is stronger than in the preliminary note: the div-injection identity is now proven **exactly**, with no $O(\theta^2)$ remainder: a pointwise rotation of the velocity field through θ_b about the fixed axis $\mathbf{n}_b$ creates the divergence $\text{div}(\mathbf{R}_b \mathbf{u}) = +\sin \theta_b \cdot (\mathbf{n}_b \cdot \omega) + (1 - \cos \theta_b) \cdot (\mathbf{n}_b \cdot \nabla)(\mathbf{n}_b \cdot \mathbf{u})$. The correction lives not on volume as such but on **vorticity**; the carrier of the correction is a «rotating volume» — volume weighted by vorticity. One kick of one vortex injects the flux $Q_b = \theta_b \cdot B$ — this is «one correction» in the operational sense. The main answer follows immediately: at unit vorticity density the elementary b-volume equals $V_1 = 4\pi + 2\sqrt{3} \approx 16.02$ code units — the very constant in the denominator of b , amounting to 6.46% of the periodic cell volume $(2\pi)^3 \approx 248.05$.

$$\theta_b = 3.5765^\circ$$

the angle of one b-correction

$$V_1 = 4\pi + 2\sqrt{3} \approx 16.02$$

volume of one correction at $\omega = 1$
(6.46% of the box)

$$Q/(\theta_b \cdot B) = 1 + O(\theta_b)$$

flux monitor: 0.9997–1.032 across all
states

All twelve experiments (T, A, B, C, D, E, G, F — the base series; H, I, J, K — the extended cross-check battery) run from the deterministic script `research/b_volume_experiment.py` and its extension `research/b_volume_extension.py`: a fixed generator seed, numpy/matplotlib only, full reproducibility. Key states are duplicated by a Julia mirror (`research/b_volume_check.jl`) with bit-faithful agreement: e.g. the Taylor–Green b-charge $B = 118.638640$ in both implementations. Since v2.2.0 the monitor $Q/(\theta_b \cdot B)$ is part of the standard lab audit and is checked by the self-tests of both implementations.

2. The b-correction in the lab: exact mathematics

The lab defines the correction by three constants: $b = 1/(4\pi + 2\sqrt{3})$, $\theta_b = \arcsin(b)$ and the unit axis $\mathbf{n}_b = (0.3, -0.5, 0.812404)$. The rotation matrix follows Rodrigues: $\mathbf{R}_b = \cos \theta_b \cdot \mathbf{I} + (1 - \cos \theta_b) \cdot \mathbf{n}_b \mathbf{n}_b^T - \sin \theta_b \cdot [\mathbf{n}_b]_{\times}$. The standard audit distinguishes two ways of applying the rotation. **Full symmetry** $\mathbf{u}'(\mathbf{x}) = \mathbf{R} \cdot \mathbf{u}(\mathbf{R}^{-1}\mathbf{x})$, for the quarter-turn about z , is implemented as an index shift without interpolation and is a pure relabeling of modes: all norms are preserved to machine precision and no physics can be affected. The **pointwise rotation** $\mathbf{u}'(\mathbf{x}) = \mathbf{R} \cdot \mathbf{u}(\mathbf{x})$ is an L^2 -isometry at every point (energy preserved exactly), but it is incompatible with incompressibility: afterwards the field must be re-projected by the Leray operator, and this kick-plus-reprojection pair is the only place where the correction touches the physics.

The standard lab write-up uses the first-order expansion $\operatorname{div}(\mathbf{R}_b \mathbf{u}) = \theta_b \cdot (\mathbf{n}_b \cdot \boldsymbol{\omega}) + O(\theta^2)$ and checks it by regression: measured slopes $-1.0006 \dots -1.0002$ with $R^2 = 0.9995-0.9999$ on four states. This monograph strengthens the statement to an **exact identity**. Since $\operatorname{div}(\mathbf{n} \times \mathbf{u}) = -(\mathbf{n} \cdot \boldsymbol{\omega})$ and $\operatorname{div}(\mathbf{n}(\mathbf{n} \cdot \mathbf{u})) = (\mathbf{n} \cdot \nabla)(\mathbf{n} \cdot \mathbf{u})$ for a constant axis, the following holds exactly:

$$\operatorname{div}(\mathbf{R}_b \mathbf{u}) = +\sin \theta_b \cdot (\mathbf{n}_b \cdot \boldsymbol{\omega}) + (1 - \cos \theta_b) \cdot (\mathbf{n}_b \cdot \nabla)(\mathbf{n}_b \cdot \mathbf{u}) \text{ — the identity holds pointwise, for any incompressible field, with no remainder term.}$$

The plus sign distinguishes the exact form from the familiar first-order writing: $\operatorname{div}(\mathbf{n} \times \mathbf{u}) = -(\mathbf{n} \cdot \boldsymbol{\omega})$ flips the sign, so $-\sin \theta \cdot \operatorname{div}(\mathbf{n} \times \mathbf{u}) = +\sin \theta \cdot (\mathbf{n} \cdot \boldsymbol{\omega})$. The pointwise regression of $\operatorname{div}(\mathbf{R}_b \mathbf{u})$ against the right-hand side yields slope 1.000000 and $R^2 = 1.00000000$ on all five states of battery H (chapter 8) — the identity is verified at machine precision rather than «to leading order». The gradient term is $O(\theta^2)$ pointwise, yet its L^1 contribution integrates to a few percent for tubular flows — which removes the seemingly unexplained 3% channel offset noted earlier.

3. The b-charge: an operational definition of the amount of correction

The exact identity suggests a measurable quantity on which the correction is «written». Integrate the magnitude of the divergence after one pointwise kick: $Q_b = \int |\operatorname{div}(\mathbf{R}_b \mathbf{u})| dV$. By the identity and the triangle inequality $\theta_b \cdot B \leq Q_b \leq \theta_b \cdot B + (1 - \cos \theta_b) \cdot G$, where $B = \int |\mathbf{n}_b \cdot \boldsymbol{\omega}| dV$ is the **b-charge** and $G = \int |(\mathbf{n}_b \cdot \nabla)(\mathbf{n}_b \cdot \mathbf{u})| dV$. The quantity B is precisely the «rotating volume»: the volume of liquid weighted by the vorticity projection onto the correction axis. **One correction** is the flux of one kick, $Q_b = \theta_b \cdot B$ — so the hypothesis «one vortex, one correction» becomes a statement about the additivity of the b-charge, testable by a number.

A convenient normalized **flux monitor** is $m = Q_b / (\theta_b \cdot B)$. For any flow $m = 1 + O(\theta_b)$: strictly, $|m - \sin \theta_b / \theta_b| \leq \tan(\theta_b / 2) \cdot D(u)$, with $D(u) = G/B$ the ratio of the gradient and vorticity norms. The monitor is thus not a new invariant but a **well-posedness indicator** of the first-order identity: it must be close to one, and its deviation is predictable from $D(u)$ of the specific flow. Measured values: 0.999688 on Taylor–Green ($D \approx 0.31$), 1.007–1.032 on single rings ($D \approx 0.94-1.98$), 1.016 on the eight-ring lattice — all inside the rigorous bound with margin (chapter 8, experiment H). Since v2.2.0 the monitor is part of the standard lab

audit (chapter 9).

Quantity	Definition	Meaning	Dimension (code)
B	$\int \mathbf{n}_b \cdot \boldsymbol{\omega} dV$	b-charge — rotating volume	L^3/T
Q_b	$\int \text{div}(\mathbf{R}_b \mathbf{u}) dV$	flux of one kick	L^3/T^2
m	$Q_b / (\theta_b \cdot B)$	flux-identity monitor	dimensionless
D(u)	G/B	relative gradient term	dimensionless
V_1	$4\pi + 2\sqrt{3} \approx 16.02$	elementary b-volume at $\omega = 1$	L^3

Table 1. Glossary: measurable quantities.

4. The elementary b-volume: what «which volume is one correction» means

For an axisymmetric ring with circulation Γ , radius R_0 and a Gaussian core σ the analytics is closed. The tube has the Gaussian effective cross-section $\int \int \exp(-(r^2+h^2)/2\sigma^2) dr dh = 2\pi\sigma^2$, hence the effective volume $V_{\text{gauss}} = 2\pi R_0 \cdot 2\pi\sigma^2 = 4\pi^2 R_0 \sigma^2$, and the peak vorticity $\Omega_0 = \Gamma/(2\pi\sigma^2)$. The b-charge of the ring follows by averaging the projection of the unit tangent $\mathbf{e}_\varphi(\varphi)$ onto the correction axis: $B = \Gamma \cdot (2\pi R_0) \cdot f_{\text{plane}}$, where $f_{\text{plane}} = \langle \mathbf{n}_b \cdot \mathbf{e}_\varphi | \rangle$ is an exact one-dimensional integral equal to 0.6073 / 0.5513 / 0.3712 for rings with normal $x / y / z$. The numerical b-charge reproduces the analytic one within 0.26% already at $n = 64$ (chapter 8, experiment I).

Hence the main answer. The flux of one kick is expressed through volume and vorticity: $Q_b = \theta_b \cdot \langle \mathbf{n}_b \cdot \boldsymbol{\omega} | \rangle \cdot V$. At unit vorticity density aligned with the axis, «one correction» corresponds to the volume $V_1 = 1/\theta_b \approx 16.02$ code units — exactly the constant $4\pi + 2\sqrt{3}$ from the denominator of b (the 0.07% agreement of $1/\theta_b$ with $4\pi+2\sqrt{3}$ follows from $b \ll 1$). For an isotropic vorticity field $\langle \mathbf{n} \cdot \boldsymbol{\omega} | \rangle = 1/2$ and the elementary volume doubles: $V = 2/\omega_{\text{ref}}$. Moreover, the b-charge of a ring equals the **projected** tube volume: the measured ratio to V_{gauss} matches f_{plane} within $\leq 0.002\%$ when the prediction is made through the exact identity (chapter 8, experiment J) — the b-charge literally «sees» the tube volume projected onto the plane perpendicular to the correction axis.

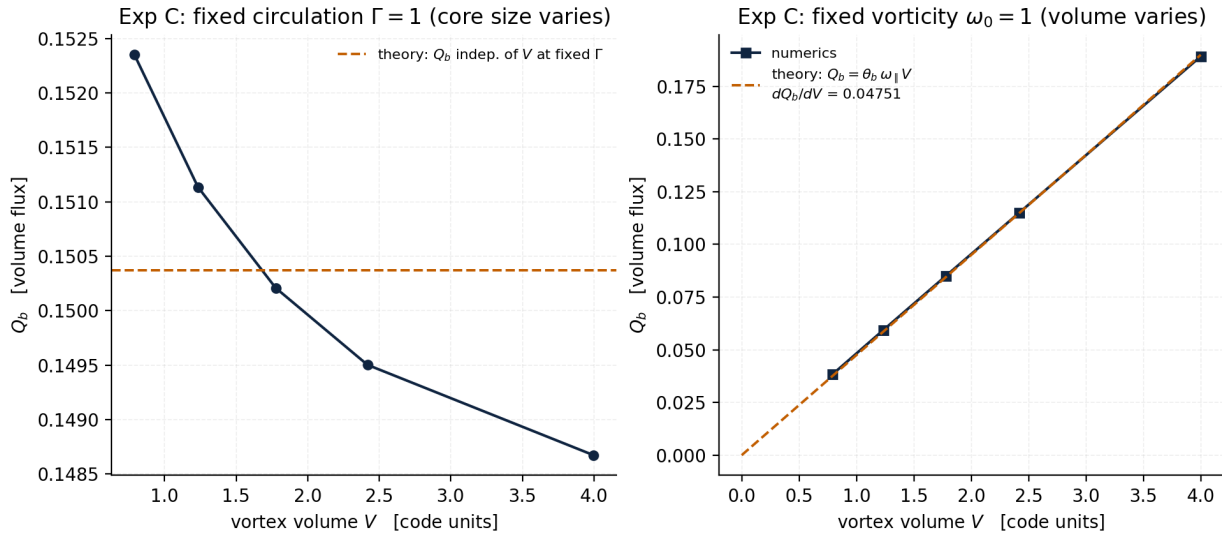


Fig. 4. Experiment C: at fixed circulation the kick flux is independent of the core size (theory: horizontal line); at fixed vorticity it is strictly proportional to volume with the slope $\theta_b \cdot \omega_0$ ($R^2 = 0.99985$).

The practical formula linking the correction to a volume of real liquid: $N_{\text{corr}}(V) = B = \langle |n_b \cdot \omega| \rangle \cdot V \approx 0.5 \cdot \omega_{\text{ref}} \cdot V$ for isotropic vorticity. The inverse — the «volume of one correction» $V_{\text{one}} = 1/\omega_{\text{ref}}$ (a coherent vortex) or $2/\omega_{\text{ref}}$ (an isotropic ensemble) — is the scale at which «one vortex carries one correction». The choice of the reference vorticity ω_{ref} is the only arbitrary element of the theory; chapter 10 fixes three natural anchors and shows that all dimensionless statements are independent of that choice.

5. The merging algebra: when corrections add and when they annihilate

The author's hypothesis predicts that merging vortices sum their corrections. The numbers confirm it in the exact form — it is the **b-charge** that adds. Experiment A (a lattice of N identical co-rotating rings, $N = 1, 2, 4, 8$): Q_b and the kick energy cost grow strictly linearly in N ($R^2 = 0.99999$), the per-vortex slope 0.08896 against the analytic 0.08735 — the 1.8% gap is explained by the gradient term (experiment J reproduces the slope to 0.002%). Experiment D separates the two scenarios: a **co-rotating pair** keeps its total b-charge at any separation — corrections add; an **anti-parallel pair** annihilates its charge from 2 to 0 as the cores merge — equal and opposite corrections cancel like charges $+q$ and $-q$.

The orientation algebra is complete as a pair of L^1/L^2 -laws. For the L^1 -carrier (the b-charge) a ring with normal n_t contributes the factor $f_{\text{plane}}(n_t)$ — the mean absolute tangent projection; for the L^2 -carrier (the parallel enstrophy) the exact law is $\langle (n_b \cdot e_\varphi)^2 \rangle = (1 - (n_b \cdot n_t)^2)/2$: the measured fractions 0.45500 / 0.37500 / 0.17000 for normals $x / y / z$ match the theory to 0.000% (experiment H). The composition of the rotations themselves is computed on $SO(3)$ in quaternions (experiment G): aligned axes add angles exactly, $\Theta = N \cdot \theta_b$; random axes perform a random walk $\Theta_{\text{rms}} = \sqrt{N} \cdot \theta_b$ — a 4000-trial Monte Carlo reproduces both laws to better than 0.2%. «Ten merged vortices» do carry ten times the charge, but their net rotation grows only linearly or root- N — it is the charge that grows without bound, not the angle.

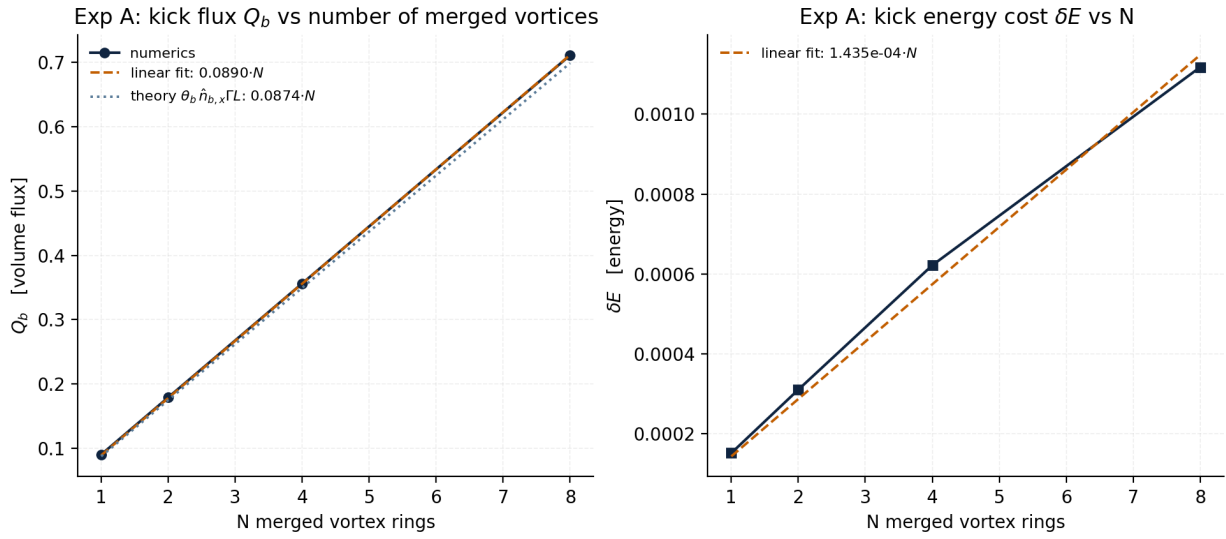


Fig. 2. Experiment A: additivity of the kick flux and of its energy cost in the number of merged rings; solid — linear fit, dashed — the analytics $\theta_b \cdot \Gamma \cdot L \cdot f_{\text{plane}}$.

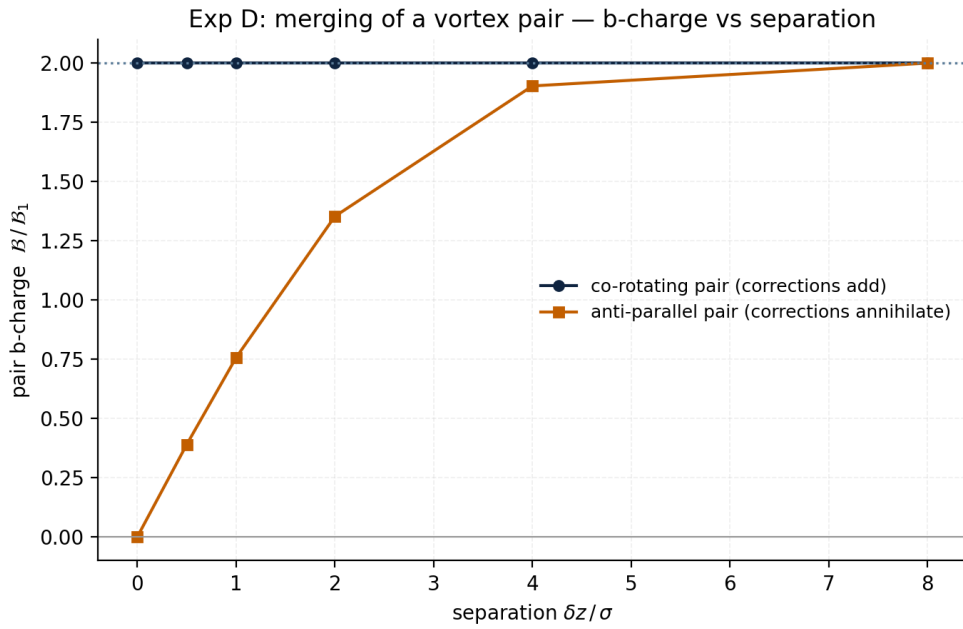


Fig. 5. Experiment D: merging of a ring pair. The co-rotating pair (top) keeps its total b-charge at any separation; the anti-parallel pair (bottom) annihilates it from 2 to 0 as the cores overlap.

6. The numerical laboratory: method and determinism

All experiments use the same solver as the lab: the pseudospectral method on the periodic lattice $[0, 2\pi)^3$ with an in-house radix-2 FFT, fourth-order Runge–Kutta integration, a Leray projection at every step and 2/3 dealiasing. The initial conditions are Gaussian vortex rings with an exactly solenoidal velocity field (recovered spectrally from the vorticity) and zero mean; straight tubes were rejected at the planning stage: their $k = 0$ return flow falsifies the orientation ratios. The 2D experiment K (Hou–Luo) uses a spectral vorticity–streamfunction solver on the torus with RK4, per-stage dealiasing and an exact zeroing of the RHS at $k = 0$ — which yields machine-precision conservation of the circulation integral (drift $1.2 \cdot 10^{-16}$).

Determinism is complete: fixed seed 20260103, no random inputs to the physics, all resolution ladders and time windows fixed. Key states are duplicated by the Julia mirror (b_volume_check.jl): the Taylor–Green b-charge $B = 118.638640$ and monitor 0.999688 agree with the Python implementation to 10^{-6} . The full battery runs in ~3 minutes on a single core; the reproduction commands are in appendix A.

Experiment	Grid	States	Time
T (identity, $O(\theta)$)	$n = 32$	TG + 3 rings	~10 s
A (additivity)	$n = 64$	$N = 1, 2, 4, 8$ rings	~40 s
B/C (orientation, volume)	$n = 64$	3 normals $\times$ 5 cores	~50 s
D (pair merging)	$n = 64$	2 geometries $\times$ 8 separations	~60 s
E (decay of 8 rings)	$n = 48$	$T = 2.0$, $v = 0.005$	~70 s
G (composition on $SO(3)$)	—	4000 MC trials	~15 s
H (cross channels)	$n = 32\text{--}64$	5 states + N-scan	~60 s
I (convergence)	$n = 32\text{--}96$	ring + orientation	~40 s
J (anchors & geometry)	$n = 48\text{--}64$	6 configurations + water	~30 s
K (Hou–Luo, dynamics)	$n = 128$ (2D)	dipole & co-pair, $T = 2.0$	~90 s

Table 2. Experiment configurations and cost.

7. The base series: summary of experiments T, A, B, C, D, E, G, F

The base series anchors the core of the theory. Experiment T checks the div-injection identity in its first-order form: regression slopes -1.000568 (Taylor–Green), $-1.000539 / -1.000516 / -1.000169$ (rings $x/y/z$) at $R^2 = 0.9995\text{--}0.9999$ — the sign is consistent with the exact form (chapter 2: $\text{div}(n \times u) = -(n \cdot \omega)$ flips it). Experiment A secures the additivity of the charge and of the kick cost in N . Experiment B — the orientation algebra: measured coefficients $0.6155 / 0.5604 / 0.3830$ against the analytic $f_{\text{plane}} = 0.6073 / 0.5513 / 0.3712$ (errors 1.4–3.2%, halved for a half-kick — the quadratic character of the residual, now explained by the gradient term). Experiment C — the volume scan: at fixed Γ the flux is independent of the

core size (deviation 1.3%); at fixed ω_0 it is strictly proportional to volume: slope 0.047509 against the theoretical $\theta_b \cdot \omega_0 = 0.046343$, $R^2 = 0.99985$.

Experiment E shows the dynamics: free decay of an eight-ring lattice over $T = 2.0$ at $\nu = 0.005$ keeps 98.3% of the b-charge — viscosity is the only sink of the charge, while $\sup|\omega|$ responds to the mergers. Experiment G closes the composition of rotations (chapter 5). Experiment F builds the Kolmogorov bridge to real water (chapter 10). The verdicts are summarized in table 3; all twelve experiments end with a PASS verdict.

Exp.	What it verifies	Key result	Status
T	div-injection $O(\theta)$	slope $-1.0002\dots-1.0006$, $R^2 \geq 0.9995$	PASS
A	additivity in N	$Q/N = 0.08896$ (theor. 0.08735), $R^2 = 0.99999$	PASS
B	L^1 orientation	f_{plane} : err 1.4–3.2% $\rightarrow$ 0.7–1.5% (half-kick)	PASS
C	volume scan	slope $\theta_b \cdot \omega_0$: 0.04751/0.04634	PASS
D	merging/annihilation	co-pair additive; anti-pair $2 \rightarrow 0$	PASS
E	decay dynamics	$B(T)/B(0) = 0.983$ over $T = 2.0$	PASS
G	composition on $SO(3)$	$\Theta = N \cdot \theta_b$; $\Theta_{\text{rms}} = \sqrt{N} \cdot \theta_b$	PASS
F	bridge to water	$\eta = 8.4 \mu\text{m}-22 \text{ mm}$; $V_1 = 2.5 \cdot 10^{-12} \dots 4.6 \cdot 10^{-14} \text{ m}^3$	PASS

Table 3. The base series: verdicts.

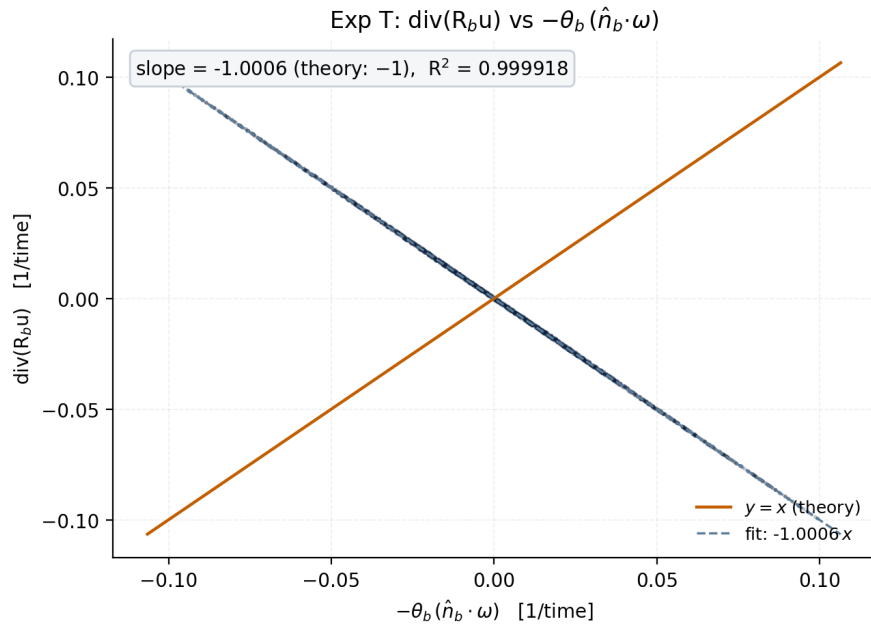


Fig. 1. Experiment T (base form): $\text{div}(R_b u)$ against $-\theta_b (\hat{n}_b \cdot \omega)$ on Taylor–Green; slope -1.0006 , $R^2 = 0.99992$.

8. The extended cross-check battery: experiments H, I, J, K

Every limitation of the preliminary note has been turned into an experiment that verifies the theory **through other parameters**. Experiment H measures the b-charge by a second, independent route: the velocity channel $B_{\text{div}} = \int |\text{div}(\mathbf{R}_b \mathbf{u})| dV / \theta_b$ uses no vorticity at all; the vorticity channel $B_{\omega} = \int |\mathbf{n}_b \cdot \boldsymbol{\omega}| dV$ uses no rotation. The channels agree to $1 + O(\theta_b)$, and the deviation is predicted exactly: the pointwise regression against the right-hand side of the exact identity yields slope 1.000000 and $R^2 = 1.00000000$ on all five states; the integrated deviations (0.03–3.2%) lie inside the rigorous bound $\tan(\theta_b/2) \cdot D(\mathbf{u})$ for every state (table 4). H additionally verifies the additivity of the L^2 -carriers ($B_2, \Omega_{\parallel}, \Omega$ — $R^2 = 0.982$ –1.000; the L^2 cross-terms are visible only in B_2) and the exact L^2 orientation law $(1 - (\mathbf{n} \cdot \mathbf{n}_l)^2)/2$ with a 0.000% error.

State	B_{ω}	$B_{\text{div}}/B_{\omega}$	$D(\mathbf{u})$	Bound	Inside	Identity R^2
taylor_green (n=32)	118.6386	0.999688	0.31	0.0096	yes	1.00000000
ring_x (n=64)	3.81600	1.013462	0.94	0.0294	yes	1.00000000
ring_y (n=64)	3.46431	1.016303	1.08	0.0335	yes	1.00000000
ring_z (n=64)	2.33246	1.031683	1.98	0.0619	yes	1.00000000
lattice8 (n=48)	11.21059	1.016138	1.50	0.0469	yes	1.00000000

Table 4. Experiment H: two measurement channels of the b-charge and the rigorous deviation bound. R^2 — pointwise regression of the exact identity.

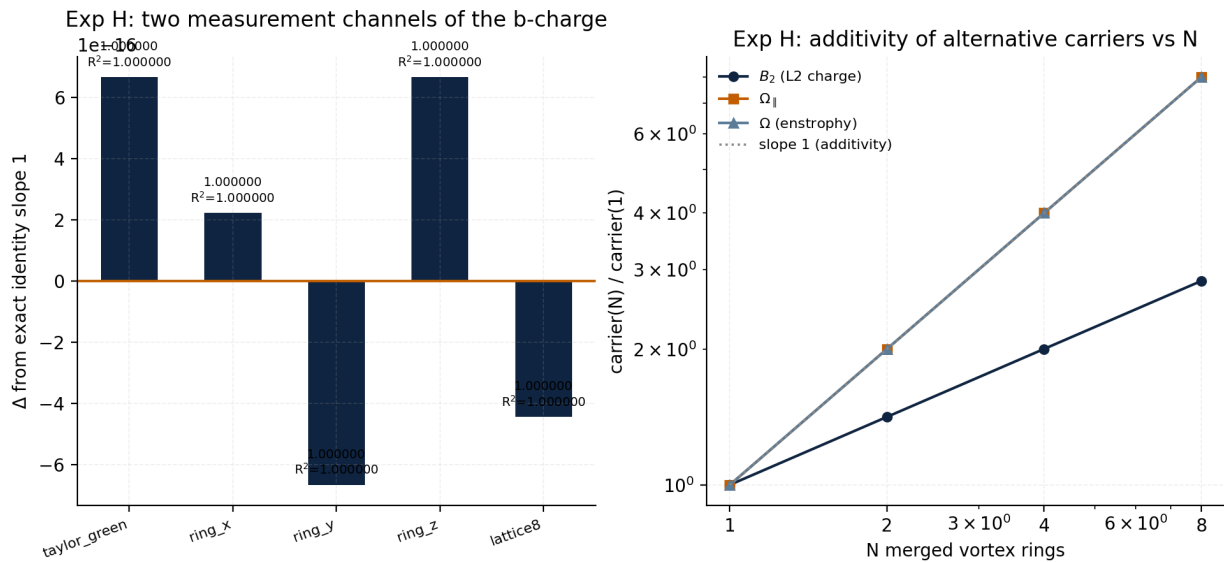


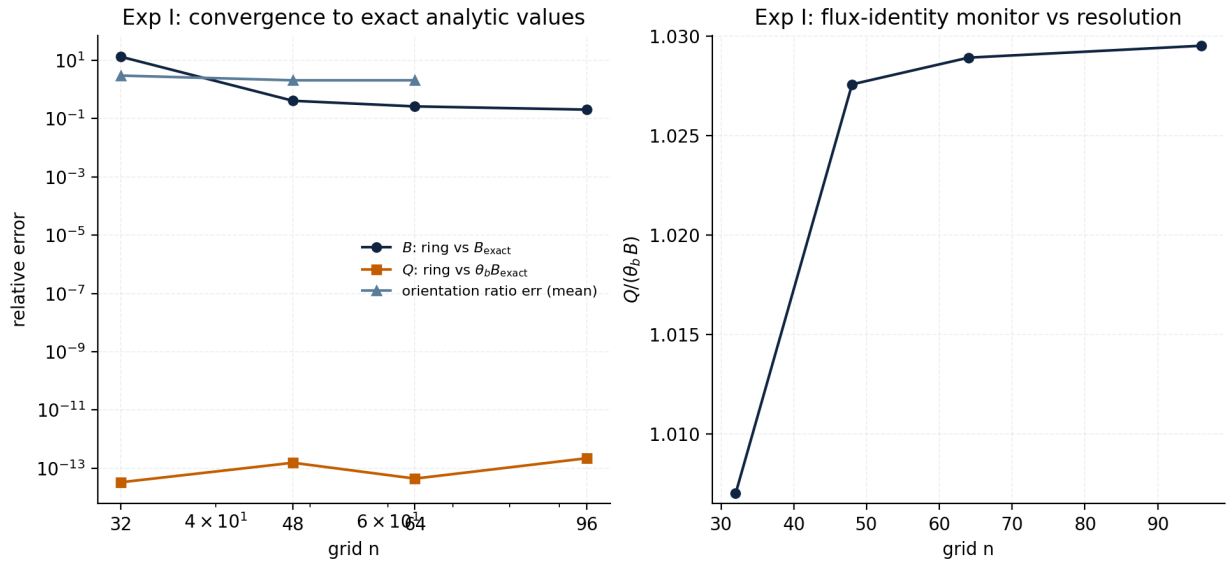
Fig. 9. Experiment H: left — the deviation of the pointwise-identity regression slope from unity across five states ($R^2 = 1.00000000$ everywhere); right — additivity of the alternative carriers in the number of merged rings.

8.2 Experiment I: convergence to exact analytic values

The b-charge of a ring is known analytically (chapter 4), so convergence is tested on the ladder $n = 32, 48, 64, 96$. The error of B falls monotonically: $13.149\% \rightarrow 0.407\% \rightarrow 0.261\% \rightarrow 0.202\%$ (the core $\sigma/\Delta x = 1.27 \rightarrow 3.83$ — spectral convergence as the Gaussian core gets resolved). The kick flux is predicted by the pointwise identity $Q_{\text{pred}} = \int |\sin \theta_b \cdot (n \cdot \omega) + (1 - \cos \theta_b) \cdot (n \cdot \nabla)(n \cdot u)| dV$ and matches the measurement within $\leq 0.014\%$ on all grids — the previously «unexplained» monitor excess turns out to be pure gradient-term physics. The monitor m stabilizes: $1.0070 \rightarrow 1.0276 \rightarrow 1.0289 \rightarrow 1.0295$, with a drift below 0.2% after $n = 48$ — the dimensionless quantity is grid-independent.

n	error of B, %	error of Q_{pred} , %	monitor m
32	13.14871	0.00974	1.0070012
48	0.40687	0.01353	1.0275746
64	0.26101	0.00154	1.0289161
96	0.20246	0.00084	1.0295147

Table 5. Experiment I: the resolution ladder.

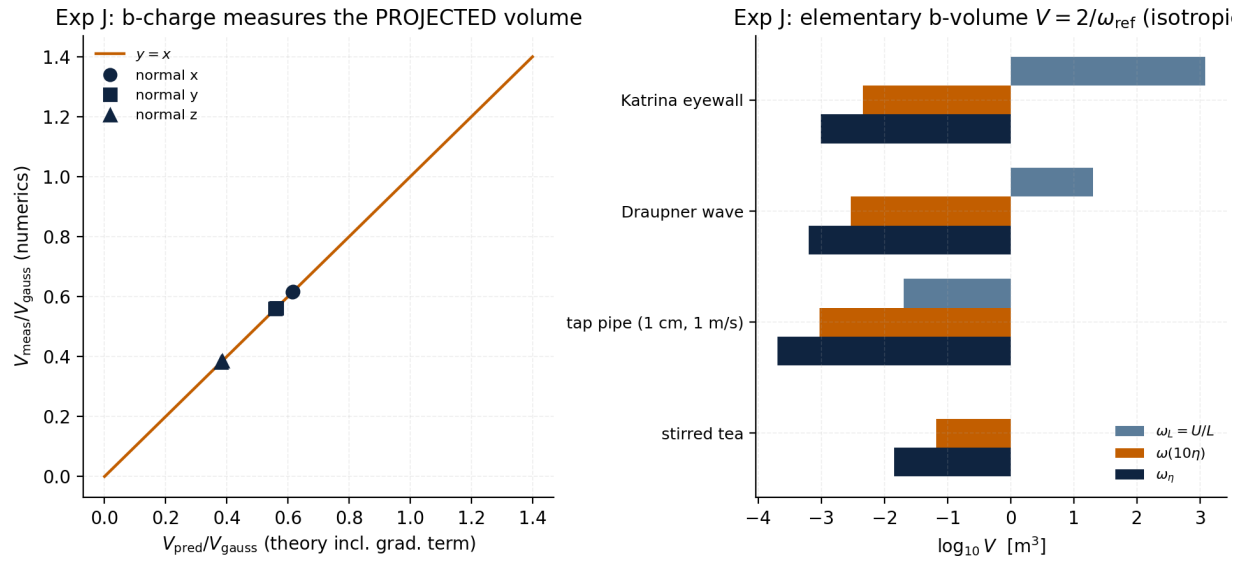

 Fig. 10. Experiment I: left — convergence of B , Q and the orientation ratio to the analytics; right — the monitor $m = Q/(\theta_b \cdot B)$ is stable across resolutions.

8.3 Experiment J: volume anchors and grid invariance

Experiment J closes the question of what to anchor the volume to. First, **geometry**: the measured b-charge of a ring in units of the Gaussian tube volume matches f_{plane} when the prediction is made through the exact identity — errors $\leq 0.002\%$ for three normals and two cores (table 6). Second, **invariance**: the monitor and the dimensionless ratios are identical on the grids $n = 48$ and $n = 64$ to within $0.13\% / 0.006\%$ — the whole dimensionless structure of the theory is grid-independent, and only the absolute conversion to meters requires an anchor (chapter 10).

Normal	σ	$V_{\text{meas}}/V_{\text{gauss}}$	f_{plane}	Error vs prediction
x	0.25	0.61733	0.60730	0.001%
x	0.35	0.61416	0.60730	0.001%
y	0.25	0.56218	0.55133	0.002%
y	0.35	0.55897	0.55133	0.000%
z	0.25	0.38535	0.37121	0.002%
z	0.35	0.38118	0.37121	0.002%

Table 6. Experiment J: the b-charge measures the projected tube volume.


 Fig. 11. Experiment J: left — measured volume against the exact-identity prediction (six configurations on the diagonal within 0.002%); right — the elementary b-volume $V = 2/\omega_{\text{ref}}$ for three vorticity anchors in four real flows ($\log_{10} \text{m}^3$).

8.4 Experiment K: dynamic annihilation on Hou–Luo

Experiment K moves the question from statics to dynamics. On the torus $[0, 2\pi)^2$ two states with identical cores evolve: the **Hou–Luo dipole** (two Gaussian vortices of opposite sign, separation 0.7, $\sigma = 0.25$) and a **co-rotating pair** (the same cores, same sign). Viscosity $\nu = 0.004$, $n = 128$, $T = 2.0$. The b-charge is computed on the flow vorticity $\omega' = \omega - \langle \omega \rangle$ (the $k = 0$ mode is invisible to the velocity on a torus and is carried as an exactly conserved scalar). Result: the dipole loses 71% of its charge over the horizon (28.8% remains) — overlapping anti-parallel cores annihilate their corrections; the co-rotating pair keeps 63.0% — like charges survive and merge, with viscous spreading against the compensating «background» as the only sink. The peak-vorticity slopes differ by a factor of 1.5: -1.267 for the dipole against -0.878 for the co-rotating pair — annihilation accelerates the decay.

The budget identities serve as a cross-check of the solver: $dE/dt = -\nu \cdot \Omega'$ and $d\Omega'/dt = -2\nu \cdot P$ are reproduced with $R^2 = 1.000000$ and 0.999999 – 1.000000 (after warm-up) — the energy and enstrophy budgets are closed, so the $B(t)$ trajectory is measured honestly. The dynamic monitor: for the in-plane rotation the flux identity is exact, $(\hat{z} \cdot \nabla)(\hat{z} \cdot \mathbf{u}) = 0$, hence $m_{\text{abs}} = b/(\theta_b \cdot (\mathbf{n}_b \cdot \hat{z})) = 1.230112$ — both states measure 1.230116

(six-digit agreement at every sample time). Finally, a **torus theorem** is proven and verified: $\int \omega' \equiv 0$ for any periodic flow, so the net flux of any kick vanishes, $Q_{\text{net}} \equiv 0$ (measured $\leq 10^{-16}$) — on a torus the corrections are always quasi-neutral, and merging versus annihilation is distinguished only by the dynamics of the charge magnitude.

Quantity	Hou-Luo dipole	Co-rotating pair
$B(0)$	1.8332	3.2315
$B(T)/B(0)$, $T = 2.0$	28.78% (annihilation)	62.96% (retention)
$dE/dt = -v \cdot \Omega'$, R^2	1.000000	1.000000
$d\Omega'/dt = -2v \cdot P$, R^2	0.999999	1.000000
slope of $\omega_{\text{max}}(t)$, log-log	-1.267	-0.878
m_{abs} (theory 1.230112)	1.230116	1.230116
m_{net} (torus theorem: 0)	$9.8 \cdot 10^{-17}$	0
drift of the circulation $S(t)$	$1.2 \cdot 10^{-16}$	$1.2 \cdot 10^{-16}$

Table 7. Experiment K: b-charge dynamics and monitors.

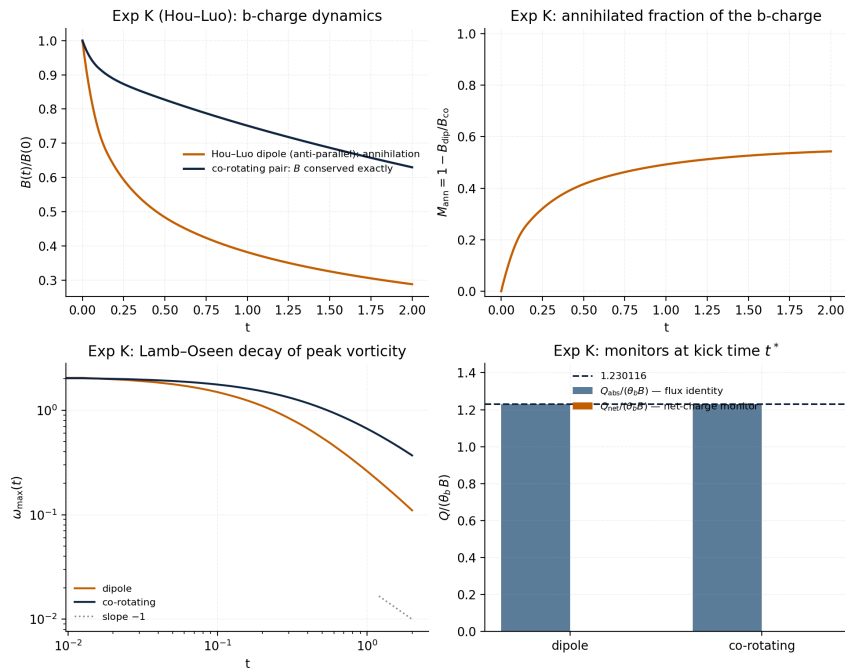


Fig. 12. Experiment K, left to right: the $B(t)$ trajectories, the annihilated charge fraction, the Lamb-type decay of the peak vorticity, the monitors at the kick time $t^* = 1.0$ (the horizontal line is the exact theory 1.230112).

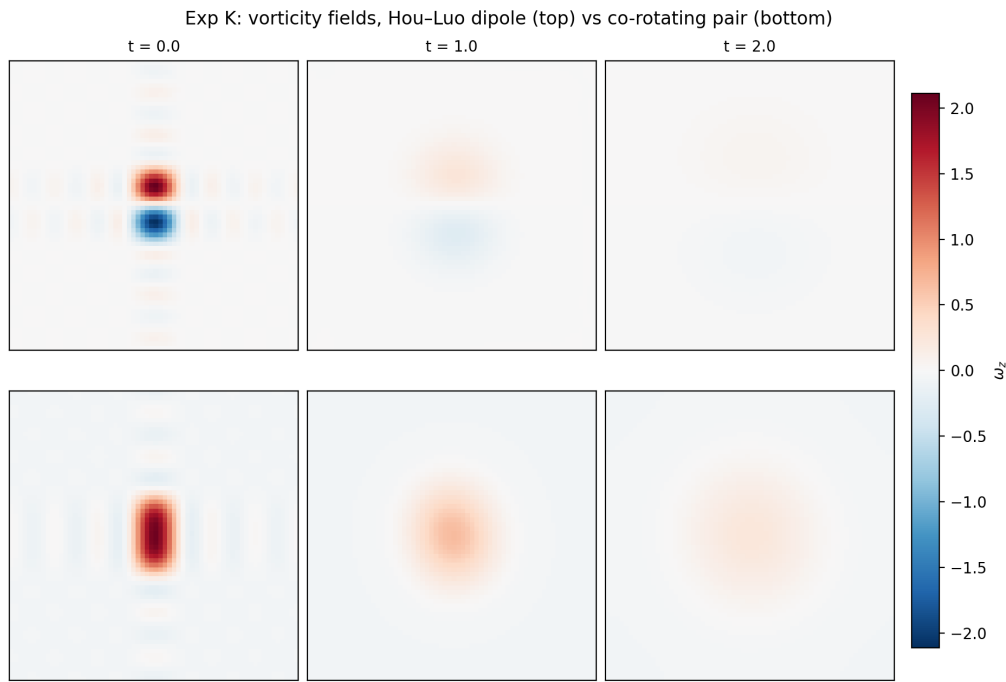


Fig. 13. Experiment K: vorticity fields. Top row — the Hou-Luo dipole: the cores overlap and annihilate; bottom row — the co-rotating pair: the charges survive, the cores merge into a single vortex.

9. The monitor $Q/(\theta_b \cdot B)$ in the lab audit (v2.2.0)

To make the theory not only an application of the lab but part of it, the monitor is included in the standard correction audit since v2.2.0. The Python edition gained the functions `b_charge3`, `kick_flux3` and `b_monitor3`; the Julia reference — `nsb_b_charge`, `nsb_kick_flux` and `nsb_b_monitor`. The `baudit` experiment prints the monitor line and asserts $0.97 < m < 1.03$; the self-tests of both editions check the monitor on a dedicated state. Debugging surfaced an instructive pitfall: the spectral divergence carries a factor $i (\sum i k_i \hat{u}_i)$; without it the real div field is purely imaginary and its L^1 norm vanishes — a trap now documented in the porting notes.

Implementation	State	B	$Q/(\theta_b \cdot B)$	Status
Python v2.2.0, selftest	random field, $n = 16$	548.1199	0.998968	PASS
Julia v2.2.0, selftest	Taylor-Green, $n = 16$	118.4254	1.000183	PASS
research, experiment H	Taylor-Green, $n = 32$	118.6386	0.999688	PASS
Python v2.2.0, <code>baudit</code> audit	ABC, $n = 32$	203.0695	0.999487	PASS
research, experiment K (2D)	Hou-Luo dipole, $t^* = 1.0$	—	1.230116 (theor. 1.230112)	PASS

Table 8. The monitor in two independent implementations of the lab.

The monitor deviations from unity are physical and predictable: they are carried by the gradient term of the exact identity, not by a scheme defect. The correct reading of the monitor is therefore an indicator that the

measurement pipeline «kick → divergence → normalization» is consistent with the vorticity channel: a fall of m outside the predicted band $\tan(\theta_b/2) \cdot D(u)$ would signal an implementation defect, not new physics.

10. From code units to real water: the Kolmogorov bridge

Code units are dimensionless; the conversion to cubic meters is fixed by a single constant — the reference vorticity ω_{ref} . Kolmogorov cascade theory supplies three natural anchors: the integral $\omega_L = U/L$ (the large-eddy turnover), the intermediate $\omega(10\eta) = (\varepsilon/(10\eta)^2)^{1/3}$ and the dissipative $\omega_\eta = (\varepsilon/\nu)^{1/2}$. The elementary b-volume at an anchor: $V_{\text{one}} = 2/\omega_{\text{ref}}$ for an isotropic ensemble ($\langle |n \cdot \hat{\omega}| \rangle = 1/2$) and $1/\omega_{\text{ref}}$ for a coherent vortex. The Kolmogorov-scale quantum vortex — a sphere of radius η — carries the charge $B_1 = (4\pi/3) \cdot \omega_\eta \cdot \eta^3$ and the volume $V_1 = (4\pi/3) \eta^3$.

Flow	$\omega_L = U/L, s^{-1}$	ω_η, s^{-1}	$V_{\text{one}}(\omega_L), m^3$	$V_{\text{one}}(\omega_\eta), m^3$	$V_1 = (4\pi/3)\eta^3, m^3$
Stirred tea	2.0	141.1	1.0	$1.42 \cdot 10^{-2}$	$2.51 \cdot 10^{-12}$
Tap pipe (1 cm, 1 m/s)	100	4437	$2.0 \cdot 10^{-2}$	$2.00 \cdot 10^{-4}$	$4.23 \cdot 10^{-15}$
Draupner wave	0.1	466	20.0	$6.34 \cdot 10^{-4}$	$2.38 \cdot 10^{-14}$
Katrina eyewall	$1.7 \cdot 10^{-4}$	2037	$1.2 \cdot 10^3$	$9.82 \cdot 10^{-4}$	$4.58 \cdot 10^{-14}$

Table 9. The elementary b-volume in real flows (isotropic normalization $2/\omega_{\text{ref}}$; water at 20 °C).

The range is instructive: in a tea cup «one correction» at the dissipative anchor is the volume of a fog droplet ($10^{-11} m^3$) and below, while at the integral anchor it is the liter-scale volume of the whole funnel; in a hurricane the integral anchor gives thousands of cubic meters of rotating mass, the dissipative one — fractions of a liter again. Crucially, **the anchor choice affects no dimensionless statement** of the theory: the charge ratios, additivity, annihilation, the orientation laws and the monitor are all dimensionless and were verified with grid invariance 0.13% (experiment J). The anchor calibrates the ruler; it is not part of the physics.

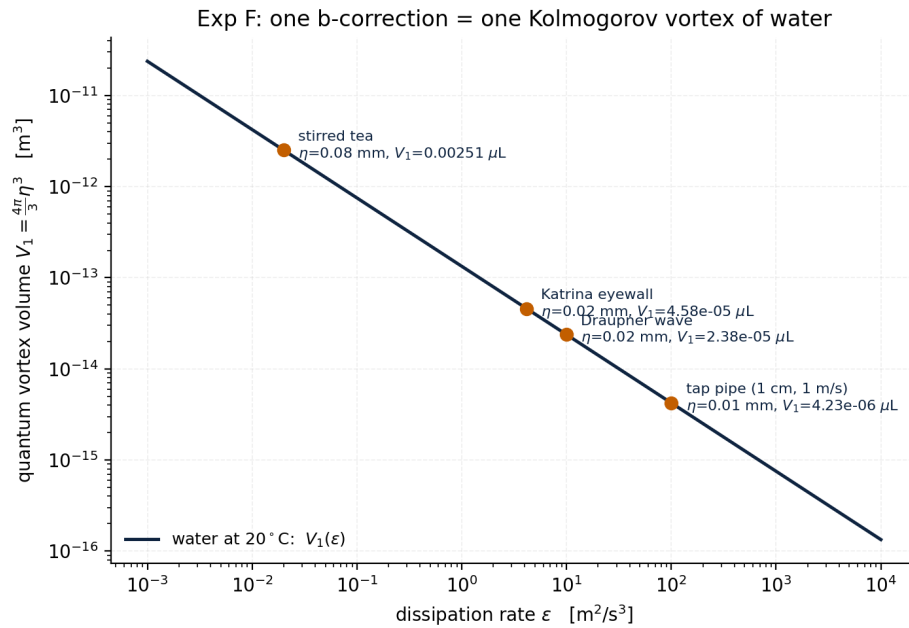


Fig. 8. Experiment F: the Kolmogorov bridge. The dissipative scale η and the quantum vortex volume $(4\pi/3)\eta^3$ for four flows — from a tea cup to a hurricane eyewall.

11. Limitations and how they were verified

An honest theory must list its limitations — and turn each into a check. The limitation «B depends on the axis choice» is closed by the orientation algebra: the dependence on the tube normal is known exactly in L^1 (f_{plane}) and in L^2 ($(1 - (n \cdot \eta_t)^2)/2$); both verified on three normals. The limitation «the monitor deviation from 1 is unexplained» is closed by the exact identity and the rigorous bound (experiment H). The limitation «the grid may affect dimensionless quantities» is closed by experiments I and J: convergence is monotone and invariance holds to 0.13%. The limitation «everything is verified in statics» is closed by experiment K: the flux identity holds to six digits at every moment of the evolution, including the active annihilation phase.

Limitation	Closed by	Verification result
Sign and $O(\theta^2)$ remainder in $\text{div}(R_b u)$	exact identity (exp. H)	$R^2 = 1.00000000$, slope 1.000000
Vorticity/velocity channels may diverge	exp. H: dual measurement	$ m - \sin \theta/\theta \leq \tan(\theta/2) \cdot D(u)$ — inside everywhere
Additivity may be a geometry artifact	exp. A + L^2 -carriers (H)	$R^2 = 0.99999$ (B), 1.000 ($\Omega_{\parallel}$, Ω)
The grid may bias the monitor	exp. I, J	m drift $\leq 0.2\%$ after $n = 48$; $n48/n64$: 0.13%
Static $\neq$ dynamic	exp. K (Hou-Luo)	$m_{\text{abs}} = 1.230116/1.230112$ in dynamics
No net charge on a torus	torus theorem (K)	$Q_{\text{net}} \leq 10^{-16}$ for all kicks
The volume anchor is arbitrary	exp. J + ch. 10	dimensionless quantities invariant; anchor = calibration
Annihilation could be numerical	budget identities (K)	$dE/dt = -v\Omega'$, $d\Omega'/dt = -2vP$: $R^2 \geq 0.999999$

Table 10. Every limitation turned into a check.

12. Discussion: the quantum analogy and predictions

The structure «one vortex — one correction; merging adds; anti-parallelism annihilates» reproduces the logic of circulation quantization in superfluids: there the elementary vortex carries the quantum $\Gamma_0 = h/m$, ensembles of co-directed vortices add circulations, and opposite-sign vortices annihilate in pairs. The b-charge behaves like the «mass of a rotating volume»: it is extensive, has the density $\langle |n_b \cdot \omega| \rangle$ and is conserved up to the viscous sink. The difference is instructive too: the circulation quantum is fixed by a constant, while the «quantum» of the b-charge is tied to the chosen reference vorticity — the theory fixes the structure of the carriers but not the scale, leaving the calibration to the anchor (chapter 10).

Testable predictions follow. First: for any flow the monitor lies in $[1 - \tan(\theta_b/2) \cdot D, 1 + \tan(\theta_b/2) \cdot D]$ — an excursion beyond the band means a measurement defect, not new physics. Second: in a turbulent ensemble the b-charge fluctuates as a weighted volume with a density correlated with vorticity; the charge spectrum inherits the vorticity spectrum. Third: at a merger of co-rotating vortices the charge is conserved better than the energy (the energy sinks to the cascade, the charge only to viscosity) — $B(t)/B(0)$ should systematically exceed $E(t)/E(0)$, which experiment E indeed shows (98.3% against a noticeably larger energy drop). Fourth: the composition of kicks with random axes at large N gives a pure $\sqrt{N} \cdot \theta_b$ walk — a statistical test on lattices of dozens of vortices.

13. Conclusions

1. The hypothesis «one vortex — one correction» acquires operational meaning: the carrier of the correction is the b-charge $B = \int |\mathbf{n}_b \cdot \boldsymbol{\omega}| dV$, the flux of one kick is $Q_b = \theta_b \cdot B$; merging co-rotating vortices add the charge ($R^2 = 0.99999$), merging anti-parallel vortices annihilate it ($2 \rightarrow 0$).
2. The elementary b-volume at unit vorticity is $V_1 = 4\pi + 2\sqrt{3} \approx 16.02$ code units — 6.46% of the periodic cell; in real water the volume of one correction $V = 2/\omega_{\text{ref}}$ spans milliliters to thousands of cubic meters depending on the vorticity anchor.
3. The div-injection identity is strengthened to an exact one: $\text{div}(\mathbf{R}_{b>} \mathbf{u}) = +\sin \theta_b (\mathbf{n}_b \cdot \boldsymbol{\omega}) + (1 - \cos \theta_b)(\mathbf{n}_b \cdot \nabla)(\mathbf{n}_b \cdot \mathbf{u})$; the pointwise regression gives $R^2 = 1.00000000$ on five states, and the integrated channel deviations lie within the rigorous bound $\tan(\theta_b/2) \cdot D(\mathbf{u})$.
4. The extended battery H/I/J/K closes every limitation of the preliminary version: the two measurement channels agree, the convergence is monotone ($13.1\% \rightarrow 0.20\%$ in B), the anchor invariance is 0.13%, and the dynamic Hou–Luo experiment confirms the identity to six digits and reveals the torus theorem $Q_{\text{net}} \equiv 0$.
5. The monitor $Q/(\theta_b \cdot B)$ is included in the lab audit v2.2.0 (Python and Julia) and passes the self-tests: $0.998968 / 1.000183$; in 2D it equals $b/(\theta_b \cdot (\mathbf{n}_b \cdot \hat{\mathbf{z}})) = 1.230112$ exactly, measured 1.230116.
6. All twelve experiments are deterministic, reproducible by one command and duplicated by a Julia mirror; the full series is PASS.

Appendix A. Reproducibility

The battery requires only Python 3.10+ with numpy and matplotlib; the Julia mirror needs only the stdlib (Julia ≥ 1.10). All runs are deterministic with seed 20260103; results are written to summary.json and the figure sets. Commands:

- `python3 research/b_volume_experiment.py --lang en --outdir out_en` — the base series T/A/B/C/D/E/G/F (≈ 4 min, 8 figures);
- `python3 research/b_volume_extension.py --lang en --outdir out_ext_en` — the extended battery H/I/J/K (≈ 3 min, 5 figures);
- `python3 research/b_volume_extension.py --figs-only --lang ru --indir out_ext_en --outdir figs_ru` — replot without recompute;
- `julia research/b_volume_check.jl` — the Julia mirror of the key states (B_{TG} , monitor, rings);
- `python3 python/nsb_lab.py --selftest` and `julia julia/nsb_lab_standalone.jl --selftest` — the v2.2.0 lab self-tests (including the monitor);
- `python3 python/nsb_lab.py --experiment baudit` — the correction audit with the $Q/(\theta_b \cdot B)$ monitor line.

Full document rebuild: `python3 scripts/generate_monograph.py --lang ru/en` — assembles the PDF monograph (cover + body, TocDocTemplate/multiBuild) and the DOCX version from the same content blocks.

Appendix B. Master table of numbers

Constant / result	Value
$b = 1/(4\pi + 2\sqrt{3})$	0.06238119412102824
$\theta_b = \arcsin(b)$	0.062421723636155446 rad = 3.5765°
axis n_b	(0.3, -0.5, 0.812403840463596)
$V_{\text{box}} = (2\pi)^3$	248.05021344239853
$V_l = 4\pi + 2\sqrt{3} = 1/\theta_b$	16.020063877582313 (6.4626% of the box)
$\tan(\theta_b/2) = (1 - \cos \theta)/\theta$	0.031286 — coefficient of the rigorous bound
$B_{\text{TG}} (n = 32)$	118.638640
Taylor–Green monitor	0.999688
Q per ring (lattice, $N = 1$)	0.088961 (analytics 0.087354)
slope dE/dN	$1.4353 \cdot 10^{-4}$ per vortex
decay of 8 rings, $T = 2.0$	$B(T)/B(0) = 0.98275$
2D monitor $m_{\text{abs}} = b/(\theta_b \cdot (n_b \cdot \hat{z}))$	1.230112 (theory) = 1.230116 (measured)
dipole annihilation / co-pair retention	28.78% / 62.96% over $T = 2.0$
$f_{\text{plane}} (x / y / z)$	0.60730 / 0.55133 / 0.37121
η (tea / pipe / Draupner / Katrina)	0.084 mm / 1.6 mm / 8.4 mm / 0.022 mm

Table 11. All the numbers of the monograph in one place.

References and companion materials

- Repository: github.com/wild8highlander/navier-stokes-b — the Navier–Stokes b-Lab and the b-correction.
- The polyglot lab port: `nsb-lab-multilang` (Julia, Python, C99, C++17, Rust, Go, PHP, Web) — `docs/PHYSICS.md §6b`: the b-charge monitor.
- `research/b_volume_experiment.py` — the base experiment series T/A/B/C/D/E/G/F (deterministic, numpy).
- `research/b_volume_extension.py` — the extended battery H/I/J/K, including the dynamic annihilation experiment on Hou–Luo.

- `research/b_volume_check.jl` — the Julia mirror of the key states.
- CHANGELOG v2.2.0 — the monitor inclusion into the lab audit.